MULTISIDED PLATFORMS (MSPs) FOR INVOLVING PATIENTS IN THE SOCIAL INNOVATION PROCESS

Patient Innovation
SHARING SOLUTIONS, IMPROVING LIFE

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SCAN ME!

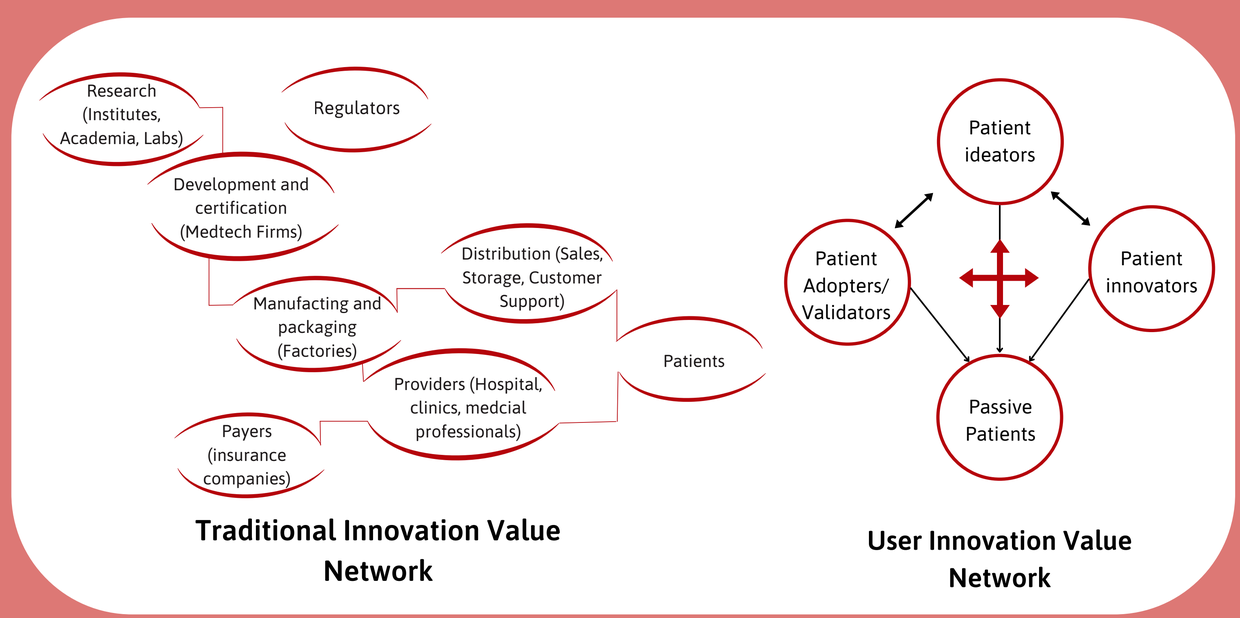
INTRODUCTION

According to a study conducted by Oliveira et al. (2015), 36% of patients and informal caregivers innovate for personal needs related to the medical condition that they face, for which the market is not providing any answer.

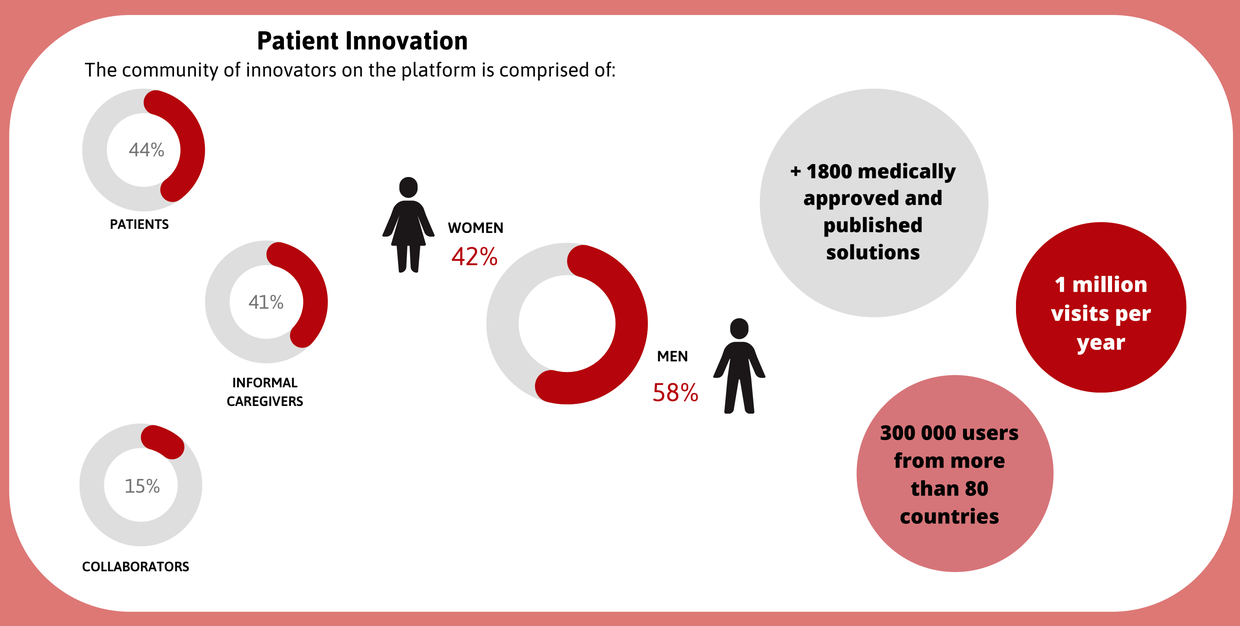
- **8%** Of patients and caregivers created solutions new to the world
- **89%** Shared their solutions with other patients
- **5%** Shared their solutions with medical professionals

Thus, in healthcare innovation, two distinct approaches emerge:

- Mainstream innovation, led by major corporations, faces complex regulations and sees the patient only as end-user
  Traditional/producer Innovation Value Network
- User innovation (Eric Von Hippel, 1988), empowers end-users, placing them at the forefront of tailored solutions, uncovering niche needs often overlooked by mainstream methods. These important innovations stay local and undiscovered. Users have trouble finding the right places to share, and complicated processes make it tough to join existing networks. These "innovative patients" create solutions to their own health needs, often relevant to others with similar needs
  User Innovation Value Network



From this landscape, Patient Innovation was born, which is a global and open nonprofit association and social network. Patients, informal caregivers, and collaborators from any disease and country can share, after medical validation, a solution they have developed to deal with a health-related limitation they are facing.



Patient Innovation's main objective is to disseminate and scale up the ability of patients to create solutions to meet their specific needs, evaluate the positive impact of these solutions on their lives, and investigate the factors that influence the development of these patient-driven innovations

To encourage user innovation in healthcare, Patient Innovation also develops the following activities and resources:

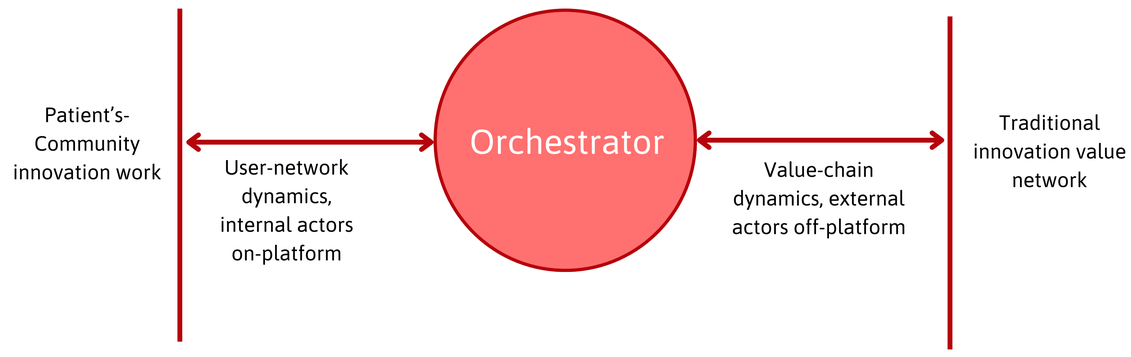
Patient Innovation Bootcamp: Boosting Patient Entrepreneurship
It is the first and only acceleration program for patients and informal caregivers, in which the goal of implementing and expanding the impact of the solutions they have developed is pursued. Through 4 editions, 44 European startups created by patients and caregivers were supported by this program with information and tools of the entire entrepreneurial journey.

Patient Innovation Awards
It is an event to give recognition and distinction to highlight the most impactful solutions, and to encourage the patient community to unleash its innovative potential. In 2023 reached its 5th edition. All editions of Patient Innovation Awards had place in big events like Web Summit, EIT Health Summit and Estoril Conferences, where 26 innovative patients, caregivers and collaborators were awarded.

PATIENT INNOVATION AS A MULTI-SIDED PLATFORM CASE-STUDY

Multi-Sided Platforms (MSPs) are new digital organizational setups that provide a proper environment to directly exchange value interactions between different types of platform users (e.g. services providers and users). (Hagiu and Spulber, 2013)

Patient Innovation, with its activities, can be considered a MSP by connecting on-platform patient's community and, at the same time, the external healthcare stakeholders.
The project guides innovative patients in the dissemination of their products, setting up startups and commercializing products, thereby unlocking patients' health innovation potential and benefiting the healthcare community and industry.
Patient Innovation highlights the crucial role of MSPs in bridging user innovators with traditional producer networks.
The objective is to leverage patient-driven innovation for improved healthcare results and nurture a genuinely patient-centered innovation ecosystem.



Patient Innovation develops its activities by orchestrating three key roles: (Cennamo et al., 2022).

Community organizer

- The platform fosters a collaborative ecossystem for patients to share their innovative solutions.

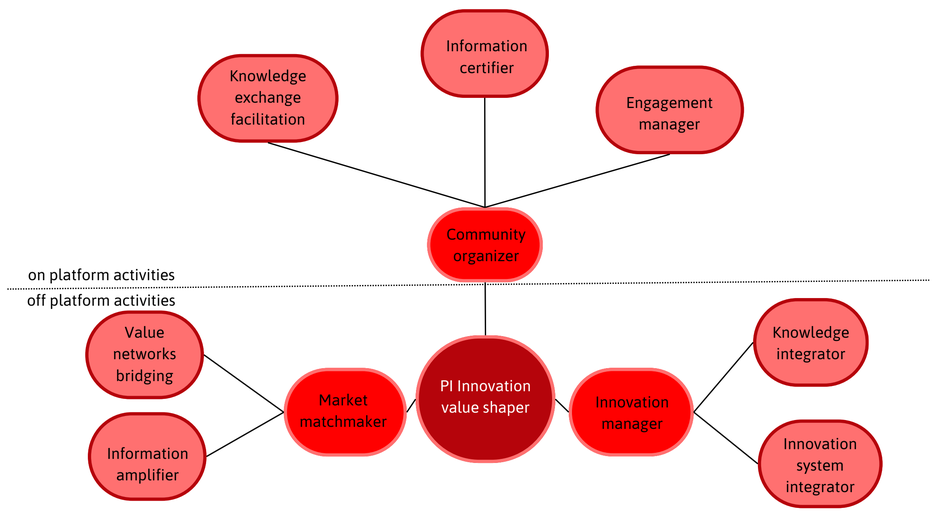
Innovation manager

- It oversees the development and commercialization processes, ensuring the transformative ideas generated by patients effectively navigate through the established producer innovation value network.

Market matchmaker

- It connects these solutions with potential market opportunities.

This multifaceted approach needs seamless interaction between on-platform activities (patient innovation sharing and collaboration) and off-platform processes (development, regulation, and market introduction).



The community organizer role encompasses three vital activities that drive innovation in both user and producer-driven value networks. By facilitating knowledge exchange, it offers a platform for users to share their innovations and provide feedback on others' solutions, thereby improving the development and dissemination of new patient innovations. This role also ensures the safety and quality of the solutions through medical evaluation, addressing user safety concerns. Moreover, the community organizer role actively promotes collaboration and engagement among users, featuring interactive buttons and solution evaluations for augmented interaction.

The Innovation Manager oversees the creation of comprehensive healthcare innovation ecosystems, guiding through regulatory challenges and providing expertise at every stage of development. This role facilitates knowledge exchange through programs like Patient Innovation Bootcamp, empowering patients to turn ideas into market-ready solutions. Additionally, it orchestrates specialized agents for efficient innovation production, allocating tasks strategically. This role optimizes patient-driven innovations, benefiting creators, end-users, and society through enhanced healthcare solutions and economic activity.

In order to enhance collaboration and ensure the widespread adoption of innovative healthcare solutions, Patient Innovation Market Matchmaker role connects patient innovators with traditional MedTech producers, and promotes high dissemination events, as Patient Innovation Awards.

CONCLUSION

It is hard for solutions developed by patients for their own needs to achieve their full potential impact on their own, as they need external validation, improvement, and integration into the healthcare supply chain in order to scale-up.
Patient Innovation aims to transform healthcare innovation by connecting users and producers. It streamlines the path from idea to market through strategic matchmaking and management, with the final goal to reshape the future of healthcare. In this sense, Patient Innovation as a MSP serves as a value shaper, bridging the gap between users and traditional producer networks, managing both on-platform and off-platform processes, involving external agents, like doctors, MedTech industry and regulators.

This raises some challenges to the platform provider, as it becomes necessary to manage the potential tensions, lack of resources and organizational and regulatory issues that might emerge. The roles described shouldn't be mutually exclusive. The most benefit will arise if they work in parallel, further enhancing the tripartite role of Patient Innovation project as a healthcare innovations value shaper. By doing this, the platform works aiming to unlock the innovation potential in healthcare that come from patients and their personal daily needs.

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